Lecture plan · for the lecturer

Sequences and the memory that fades

University, 3

Time
45 min

Materials
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

Learning goals

- The state is the memory
- Empty steps still multiply it

Activity: Predict, then slide

1. Predict (5 min): Before anyone opens the lab, pairs write the end number they expect with the memory at 0.1 and at 0.9.
2. By hand (10 min): With a cue of 1 and a memory of 0.6, pairs multiply the state through the three empty steps and write each value.
3. Lab (15 min): One phone between two: turn the memory down to “Almost none” and read the end, then raise it towards 0.95. Students correct their predictions.
4. Name it (10 min): Only now name vanishing gradients, and ask what a longer gap would need.
5. Exit (5 min): The exit ticket: does the end still know the cue? Write the number.

Discussion

- What would a longer gap do?

Exit ticket

- Does the end still know the cue? Write the number.

Try it in your next lecture
Ask the students to predict the end number before they move the memory slider.

Seminar sheet · for students

Sequences and the memory that fades

University, 3

Name _____ Group _____ Date _____

1 **In the lab "A memory that fades", the cue puts 1 into the hidden state, and the memory is 0.5. What is left at the end, after the three empty steps?**

◯ 0.25 ◯ 0.5 ◯ 0.125

2 **A recurrent net reads the whole sequence in one look.**

◯ True ◯ False

3 **Match the word to the meaning.**

Hidden state	....	The symbol that must still matter at the end
Cue	....	A small memory times the state, again and again
Fading	....	What the net carries forward

4 **Order one recurrent step.**

☐ Keep the mix as the new state

☐ Read the next symbol

☐ Mix it with the old hidden state

Answers · for the lecturer

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1 **0.125**

Each empty step multiplies the state by 0.5: 1 → 0.5 → 0.25 → 0.125.

2 **False**

It reads one symbol at a time and hopes the hidden state kept the rest.

3 **Hidden state — What the net carries forward**
Cue — The symbol that must still matter at the end
Fading — A small memory times the state, again and again

The lab's end number is the cue after it has been multiplied by the memory at every empty step.

4 **Top to bottom: 3, 1, 2**

The new state is the only memory the next symbol will see.

For you to think about

What sentence will you reject: "it reads the whole sequence at once"?